Supply Chain Performance Optimization: Lumber Delivery Predictive Analytics

By Raymond Ding

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Objectives & Problem Statement

Objective: Analyze historical lumber delivery data (2022-2025) and economic indicators to identify key factors influencing delivery rescheduling.

Problem: Inefficiencies in truck allocation and delivery rescheduling negatively impact operational costs and customer satisfaction.

Aim:

- Understand the primary drivers of the Reschedule_Rate.
- Develop a predictive model to estimate reschedule risk.
- Provide data-driven insights to optimize planning and resource allocation.

Data Overview

Source: DELIVERY_TRACKDB.csv

Content: 763 daily records (Jan 2022 - Mar 2025) detailing lumber deliveries.

Key Variables:

- Expected vs. Actual Truck Counts (Types A, B, C)
- Outside Extra Trucks (Type A)
- Delivery Sizes (Large/Small, LBS/QTY)
- Reschedules (LBS/QTY)

Initial Feature Engineering

- **Initial KPIs Derived:**
  - **Reschedule_Rate** (Qty of Reschedules/(Large Deliveries + Small Deliveries))
  - Truck_Utilization_Efficiency
  - Delivery_Fulfillment_Rate
  - Truck Accuracies (A, B, C)
  - Average_Delivery_Size
  - Trend/Seasonal/Residual Reschedule Rate

Supplementary Data

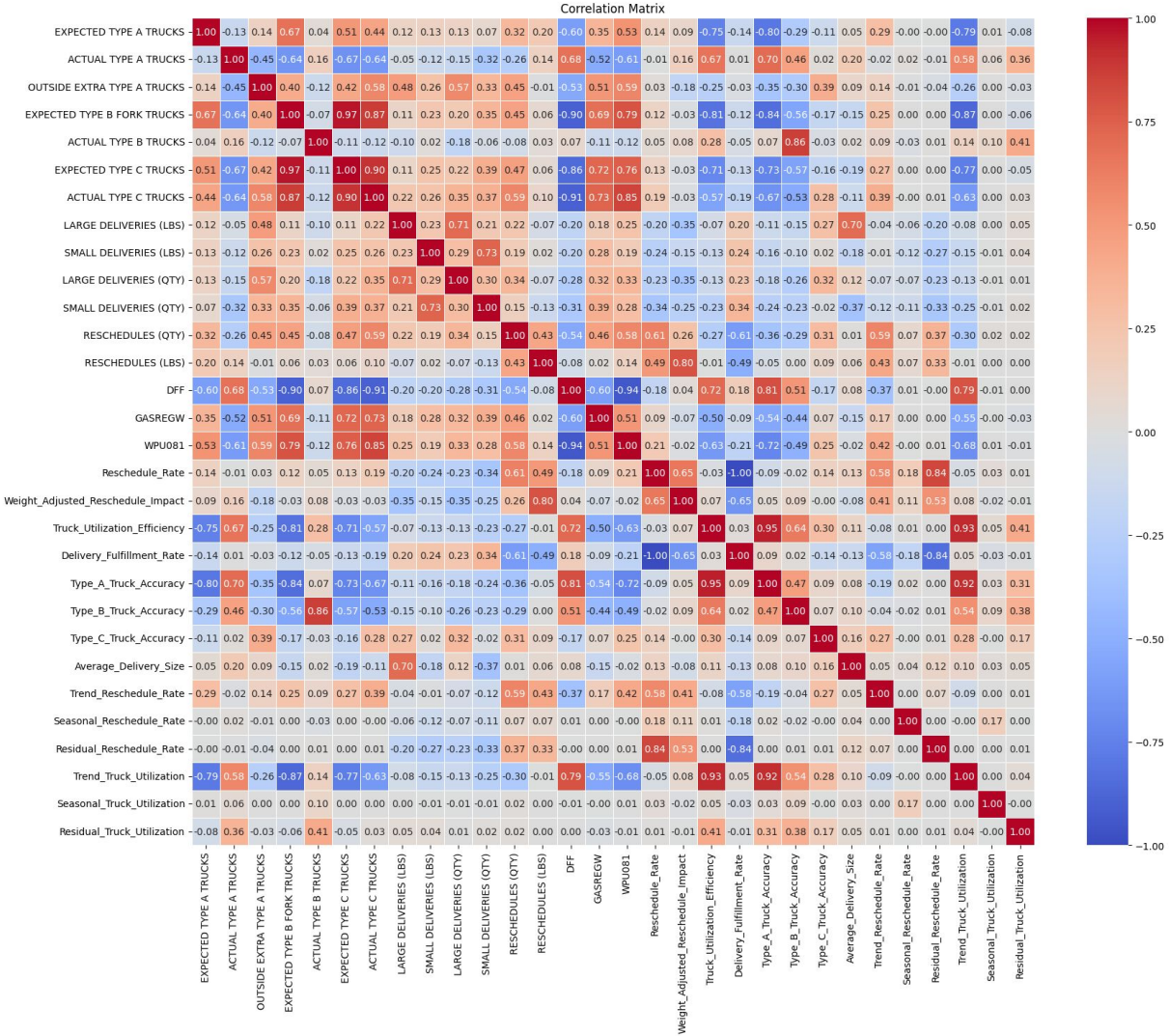
Source: Federal Reserve Economic Data (FRED)

Indicators:

- DFF: Federal Funds Effective Rate (%)
- GASREGW: US Regular Gas Price ($/gallon)
- WPU081: Producer Price Index - Lumber

Integration: Aligned daily FRED data to the business days present in the delivery dataset using forward-filling

Correlation Matrix



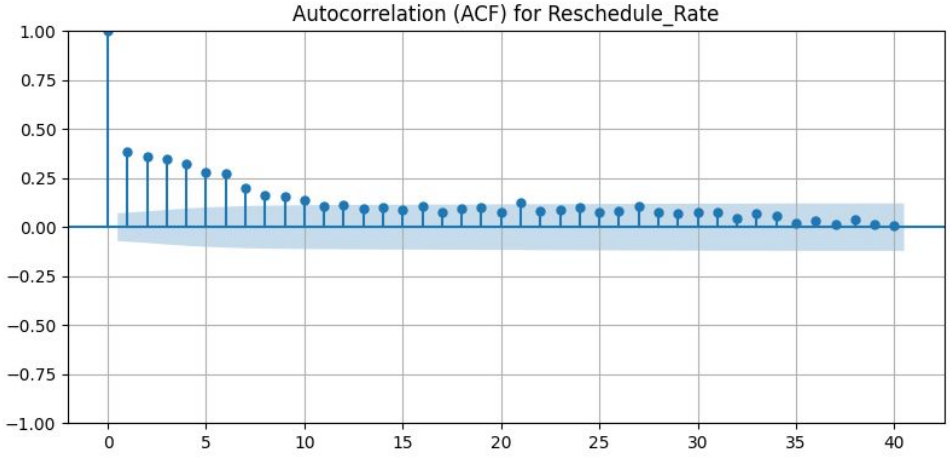
Stationarity and Feature Engineering

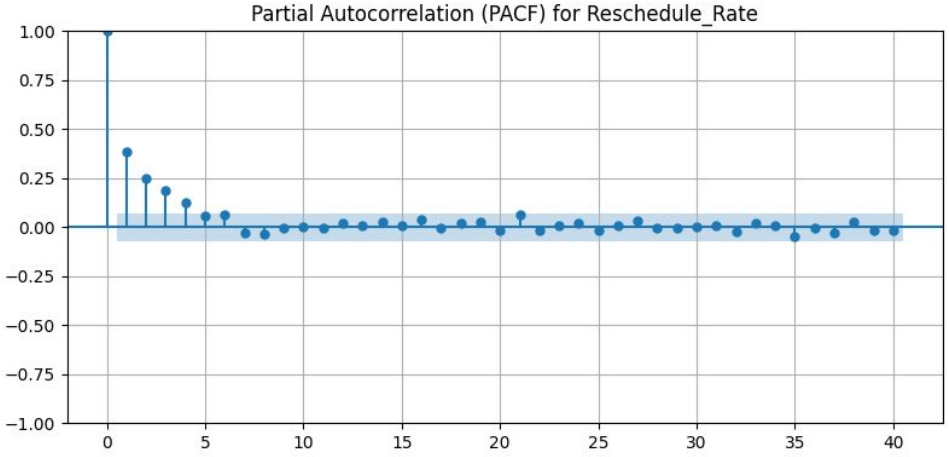
Stationarity: Used ADF and KPSS tests on all potential predictor variables.

Differencing: Differenced variables that failed one or both tests, only needed to difference once at most.

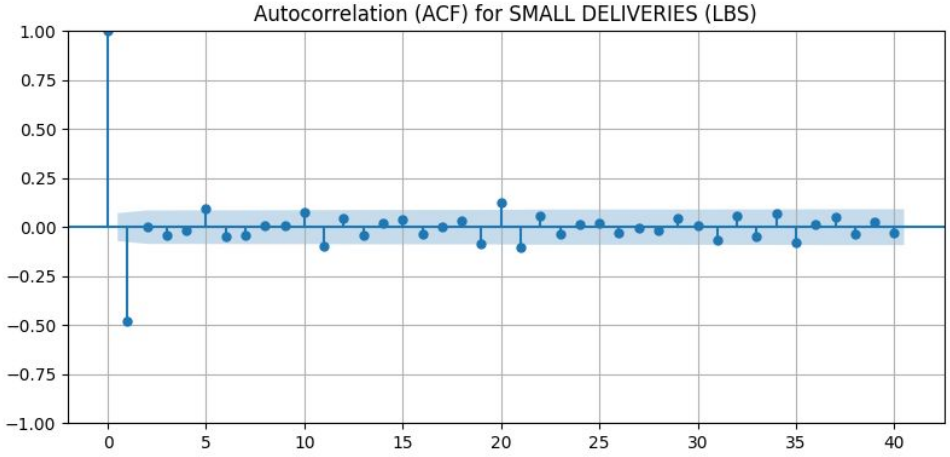
Autocorrelation: Graphed ACF and PACF graphs for each variables, then created lags based on significant lags of each variable

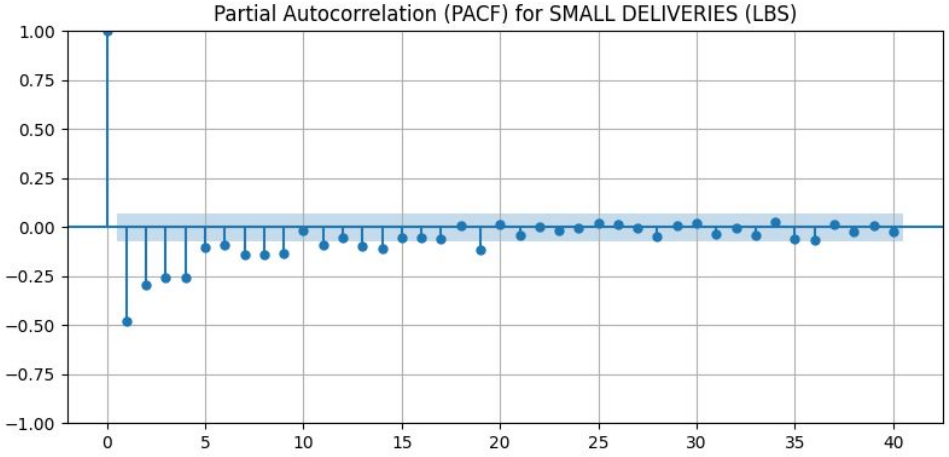
Example ACF/PACF Graph





Example ACF/PACF Graph 2





Modeling & Validation

Target Variable: Reschedule_Rate (%)

Model Choice: Elastic Net Regression

- Handles potential multicollinearity common in time series with lags.
- Performs automatic feature selection (L1/Lasso component).
- Provides interpretable coefficients.

Modeling & Validation (Cont.)

Feature Scaling: Applied StandardScaler to all features before modeling.

Hyperparameter Tuning: Used ElasticNetCV with TimeSeriesSplit (5 folds) on the training data to find the optimal regularization strength (alpha) and L1/L2 mix (l1_ratio).

Validation Strategy:

- Split data chronologically: Training Set (first 80%), Test Set (last 20%)
- Trained final model ONLY on the Training Set
- Evaluated performance metrics (R^2, RMSE, MAPE) on the unseen Test Set.

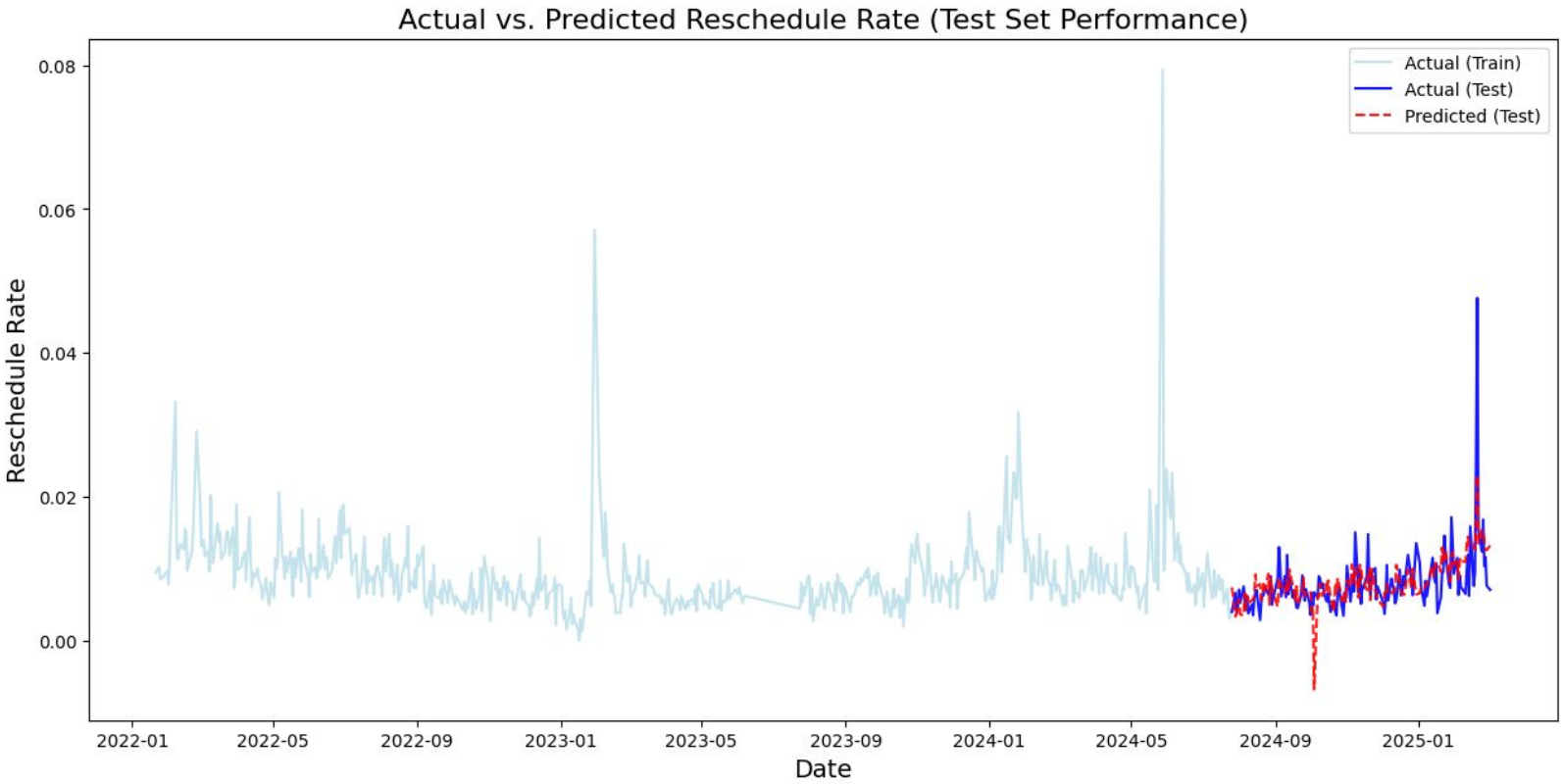
In-Sample Results

- **Best alpha: 0.00066**
  - Small alpha suggests only moderate penalty needed to optimize performance
- **Best l1_ratio: 0.14**
  - Indicates optimal mix leaned more towards the L2 (Ridge) penalty, suggesting shrinking coefficients was slightly more favored
- **R^2 score: 0.54**
  - Explains approximately 54% of the variance in the Reschedule_Rate within the training set.

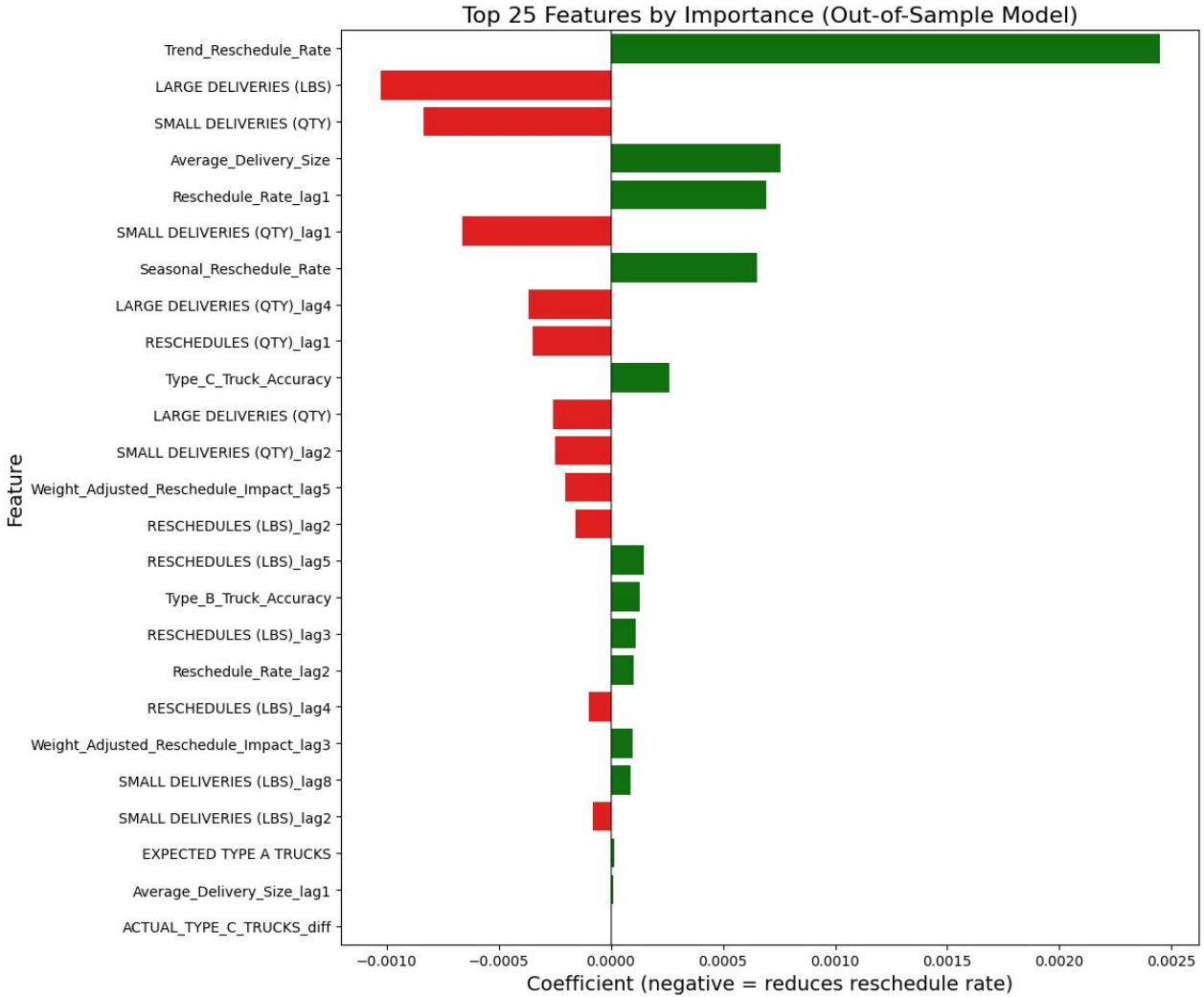
Out of Sample Results

- **R^2 Score : 0.4110**
- **Best alpha: 0.0028**
- **Best l1_ratio: 0.059**
- **Number of non-zero coefficients: 25**

Model Visualization



Feature Importance



Top Drivers - Increases Reschedule Rate

- **Trend_Reschedule_Rate (+0.0024):** The underlying long-term trend is the strongest factor. Indicates persistent systemic influences.
- **Average_Delivery_Size (+0.0008):** Larger average delivery weights are associated with higher rescheduling risk.
- **Reschedule_Rate_lag1 (+0.0007):** Strong autoregressive effect – high rates yesterday lead to high rates today.
- **Seasonal_Reschedule_Rate (+0.0007):** Identifiable seasonal patterns contribute to rescheduling.
- **Type_C_Truck_Accuracy (+0.0003):** Higher accuracy for Type C trucks paradoxically linked to more rescheduling (needs investigation).
- **RESCHEDULES (LBS)_lag5 (+0.0001):** Weight rescheduled 5 days prior has a small positive impact

Top Drivers - Decreases Reschedule Rate

- **LARGE DELIVERIES (LBS) (-0.0010):** Higher total weight of large deliveries associated with lower rescheduling.
- **SMALL DELIVERIES (QTY) (-0.0008):** Higher quantity of small deliveries strongly associated with lower rescheduling.
- **SMALL DELIVERIES (QTY)_lag1 (-0.0007):** Effect persists from the previous day.
- **LARGE DELIVERIES (QTY)_lag4 (-0.0004):** Quantity of large deliveries 4 days prior has a negative impact.
- **RESCHEDULES (QTY)_lag1 (-0.0003):** Higher quantity rescheduled yesterday slightly reduces today's rate (potential correction effect?).

(Note: Economic indicators like DFF_diff did not appear in the final Top 25 features for this model run)

Business Recommendations

Focus on High-Risk Delivery Profiles:

- **Insight:** High Average_Delivery_Size increases risk, while high SMALL DELIVERIES (QTY) decreases it.
- **Action:** Analyze processes for deliveries contributing most to high average size. Implement targeted planning checks or allocate buffer resources for these potentially complex deliveries. Promote efficiency in handling numerous small deliveries.

Leverage Predictive Insights for Planning:

- **Insight:** Trend_Reschedule_Rate and Reschedule_Rate_lag1 are strong predictors. Model provides a validated risk signal (R^2=0.41).
- **Action:** Integrate model predictions into daily planning dashboards as an Early Warning Indicator. Flag days with predicted rates above a defined threshold for proactive review, schedule confirmation, and resource readiness checks. Monitor the underlying trend closely.

Business Recommendations (Cont.)

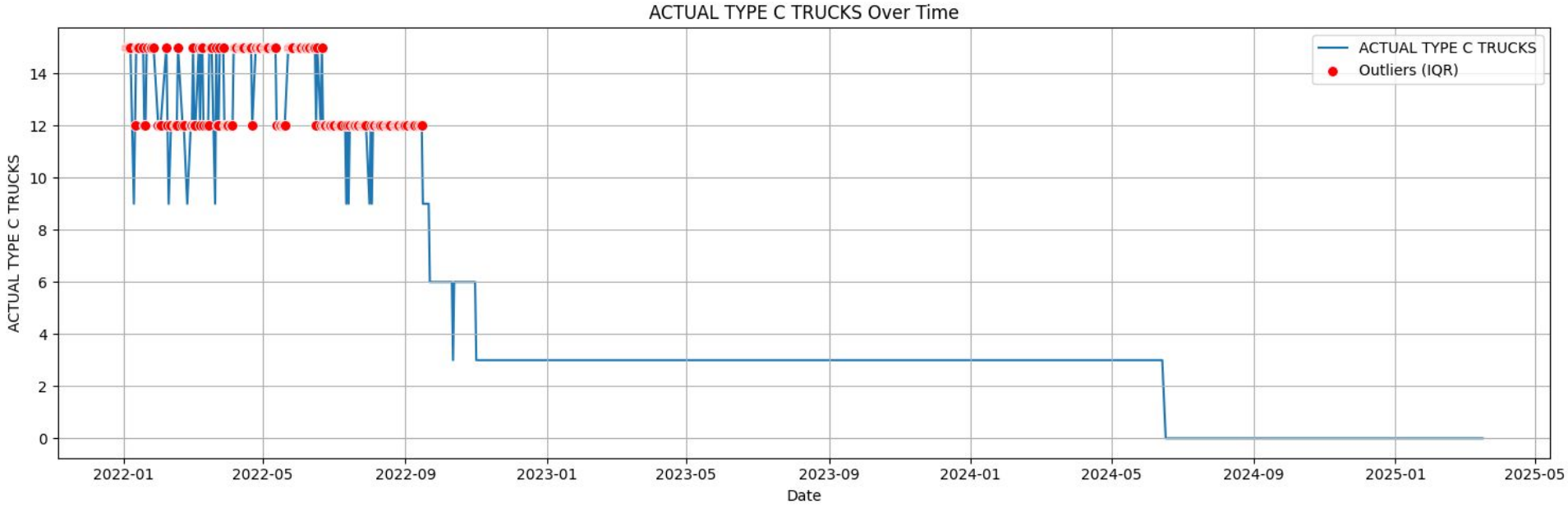
Investigate Type C Truck Operations:

- **Insight:** Positive correlation between Type_C_Truck_Accuracy and rescheduling is unusual. EDA noted Type C trucks being phased out.
- **Action:** Conduct a focused review of Type C truck usage, maintenance, and the types of jobs they handle to understand this relationship. Is it related to the phase-out or specific operational challenges?

Account for Seasonality:

- **Insight:** Seasonal_Reschedule_Rate is a significant driver.
- **Action:** Ensure operational plans explicitly incorporate known seasonal patterns in rescheduling. Adjust resource levels and customer communication strategies during historically high-risk periods.

Truck C Visualization



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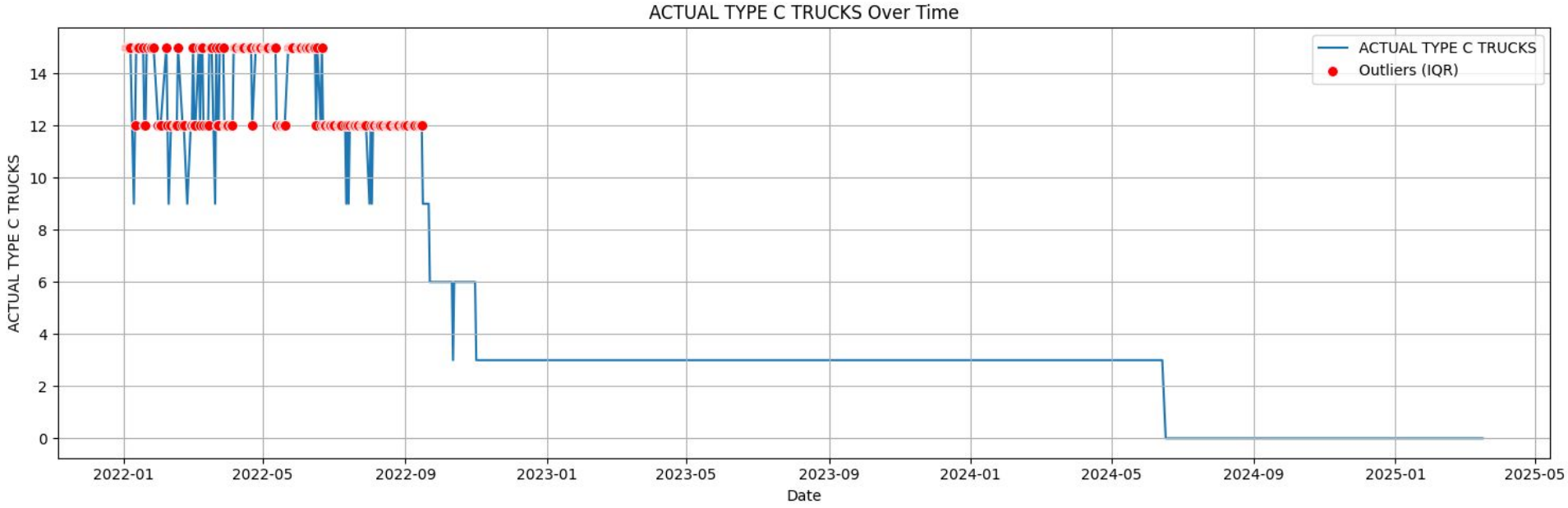
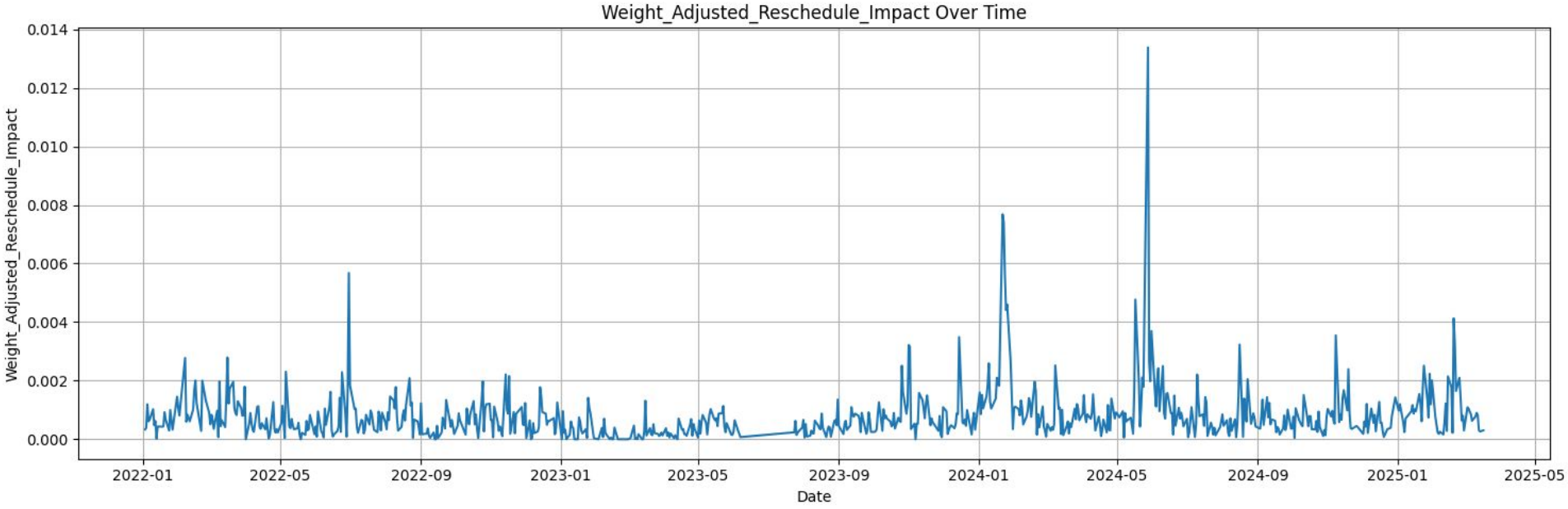


Weighted Adjusted Reschedule Impact Over Time



Limitations

Predictive Power: Model explains 41% of test set variance; significant factors remain uncaptured (e.g., weather, traffic, customer specifics, internal disruptions).

Linearity: Elastic Net assumes linear relationships; complex interactions are not modeled.

Data Granularity: Daily data might miss intra-day factors influencing rescheduling.

Stability: Model assumes relationships learned in the training period (2022-mid 2024) hold for the test period (mid 2024-early 2025). Future process changes could impact accuracy.

Future Work

Enhance Features: Incorporate external data (weather, holidays), create interaction terms, explore different lag structures.

Explore Advanced Models: Test non-linear models (Random Forest, Gradient Boosting) or time-series specific models (SARIMAX) to potentially capture more complex patterns and improve accuracy.

Analyze Other KPIs: Apply the modeling framework to predict Truck_Utilization_Efficiency or other relevant metrics.

Outlier Impact Study: Investigate the influence of extreme values on model results.

Deployment & Monitoring: Develop a strategy for integrating predictions into workflows and continuously monitoring model performance over time.

Conclusion

Conclusion: Rescheduling is driven by operational momentum (trends, lags), delivery characteristics (size/quantity profiles), and seasonal factors. A validated Elastic Net model provides meaningful insights and moderate predictive power (R^2=0.41 OOS) for risk assessment.

Key Takeaway: Data-driven insights can guide proactive planning, focusing attention on high-risk delivery types and leveraging past performance trends to anticipate future rescheduling challenges.

Recommendations: Focus on process improvements for high-risk deliveries, use the model as an early warning tool, investigate Type C truck anomalies, and continue refining the model with more data/features.

GitHub Link:

https://github.com/RaySaysHi/Lumber-Project